

Robust Differentially Private Consensus: The Effects of Noise Distributions and Lossy Communication

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Abstract—This paper investigates the robustness and performance of differentially private (DP) consensus under realistic conditions. Building on the Synchronous Private Consensus (SPC) framework, we first validate its privacy–accuracy tradeoff, confirming that stronger privacy guarantees lead to higher steady-state error. We then evaluate SPC under lossy communication. Our results show that SPC is robust to packet loss, with only moderate degradation in convergence speed and accuracy, while communication noise significantly slows convergence and increases error. To improve performance, we explore two extensions. Replacing Laplace noise with Gaussian noise (under comparable privacy guarantees) yields faster convergence, lower variance, and improved accuracy. We also propose a topology-aware noise mechanism that scales noise by node degree, improving accuracy in heterogeneous networks but introducing non-uniform privacy guarantees. Overall, our results highlight key tradeoffs between privacy, accuracy, and convergence, and demonstrate that DP consensus can remain effective in realistic settings.

Index Terms—multi-robot systems, distributed consensus, differential privacy, lossy communication

I. INTRODUCTION

Our work is at the intersection of distributed consensus, a major theme in COMPSCI 2860, and differential privacy (DP). We study DP consensus, in which a network of agents seeks to converge to a common value while ensuring that each agent’s initial state remains private.

We build on the Synchronous Private Consensus (SPC) framework [1], which achieves DP by injecting noise into agent communications while preserving probabilistic convergence. While this framework provides strong theoretical guarantees, it relies on idealized assumptions, including reliable communication and specific noise distributions.

We investigate the robustness of the SPC framework under realistic communication conditions, including channel noise and packet loss. Experimental results show that SPC remains surprisingly robust under lossy communication, with only modest degradation in convergence and accuracy under equivalent privacy guarantees. We then explore mechanisms to improve performance, measured in terms of model accuracy and convergence time, through alternative noise distributions and topology-aware communication strategies that exploit the structure of the underlying graph. Our goal is to extend the SPC framework to more realistic settings while preserving its core privacy, accuracy, and convergence guarantees.

II. LITERATURE REVIEW

A. Distributed Consensus

Consensus algorithms are the foundation of distributed multi-agent systems, enabling agents to iteratively exchange information and converge to a common value. Classical frameworks [2] model agent updates as weighted averages over local neighborhoods. Under standard connectivity assumptions, these methods guarantee convergence to the degree-weighted average of initial states [3] and have been widely applied in settings such as sensor fusion [4], load balancing [5], and multi-robot coordination [6]. These approaches assume ideal conditions, including reliable communication and full observability of agent states. They do not account for adversarial behavior or the possibility that agent states are sensitive data.

B. Differential Privacy

Differential privacy (DP) is a formal framework for protecting individual data by ensuring that the output of a computation is insensitive to any single agent’s input [7], [8]. Intuitively, DP guarantees that the outputs of two neighboring worlds (one including a given agent’s data and one excluding it) remain statistically indistinguishable, limiting an adversary’s ability to infer participation. Standard mechanisms, like the Laplace and Gaussian ones, achieve this guarantee by adding calibrated random noise based on query sensitivity [7], providing robustness against inference attacks [9].

Despite these strong guarantees, DP is primarily developed in the context of centralized data analysis, where a trusted curator releases aggregate statistics. As a result, this framework does not directly address distributed or iterative settings.

C. DP Consensus

[10], [11] considered consensus in unreliable networks with adversarial agents, focusing on detecting and identifying faulty or malicious nodes. These works found that resilience depends on graph connectivity and provided conditions under which adversaries can be detected. However, this approach assumed adversaries actively disrupt the system and focused on consensus accuracy rather than protecting the privacy of agent states. In contrast, our work considers an *honest-but-curious* adversary and aims to make agent data unidentifiable.

[1] added privacy to consensus in the Synchronous Private Consensus (SPC) framework. The mechanism adds geometrically decaying Laplace noise to agent messages, similar to other noise-based privacy-preserving methods [12], ensuring DP and probabilistic convergence. It introduced both centralized and decentralized versions, where agents in the decentralized setting communicate only with their neighbors and update using noisy local averages. The work also showed a clear tradeoff between privacy and accuracy, while preserving probabilistic convergence to consensus.

D. Limitations

While SPC provides strong theoretical guarantees, it relies on several idealized assumptions. In particular, it assumes reliable communication and does not account for communication noise or packet loss. Additionally, it used Laplace noise [13] and did not explore alternative distributions.

Together, these works highlight a gap between theoretical privacy guarantees and practical distributed systems. Existing approaches either ensure convergence without privacy, provide privacy under idealized assumptions, or address adversarial robustness without protecting data. Our work builds on these foundations to study DP consensus under more realistic lossy communication and to improve performance through alternative noise distributions and topology-aware mechanisms.

III. PROBLEM STATEMENT

A. Distributed Consensus

Consider a network of N agents, where each agent $i \in \{1, \dots, N\}$ has initial state $\theta_i(0) \in \mathbb{R}$. Let the global state at time t be $\theta(t) = [\theta_1(t), \dots, \theta_N(t)]^\top$.

Agents iteratively update states via local communication:

$$\theta(t+1) = W\theta(t), \quad (1)$$

where $W \in \mathbb{R}^{N \times N}$ is the communication topology matrix. The goal is to achieve *consensus* between agent states:

$$\lim_{t \rightarrow \infty} |\theta_i(t) - \theta_j(t)| = 0 \quad \forall i, j \in [N]. \quad (2)$$

In the centralized communication case, the desired convergence state is the *simple average of initial states*.

$$\lim_{t \rightarrow \infty} \theta_i(t) = \frac{1}{N} \sum_{k=1}^N \theta_k(0) \quad (3)$$

In the decentralized case where all nodes directly communicate with their neighbors, the desired converged state is the *degree-weighted average of initial states*.

$$\lim_{t \rightarrow \infty} \theta_i(t) = \frac{\sum_{k=1}^N (d_k + 1) \theta_k(0)}{\sum_{k=1}^N (d_k + 1)} \quad (4)$$

However, under additional conditions on the communication topology, decentralized consensus can instead converge to the simple average of the initial states [2]. In particular, convergence to the simple-average is guaranteed when the communication topology is *balanced*, i.e.,

$$|d_{\text{in}}(i)| = |d_{\text{out}}(i)| \quad \forall i \in [N]. \quad (5)$$

Having an undirected communication topology is a sufficient, but not necessary, condition for balance.

In many applications, the initial states $\theta_i(0)$ are sensitive. However, standard consensus protocols require agents to exchange information derived from these values over multiple rounds, potentially leaking private information [1] [14].

We consider an *honest-but-curious* adversary who simply observes all communication and attempts to infer the agents' initial states. The challenge is to design a consensus mechanism that ensures convergence while preventing the adversary from inferring sensitive initial states.

B. Differential Privacy

Two initial state vectors $\theta(0), \theta'(0) \in \mathbb{R}^N$ are δ -adjacent if they differ by δ in at most one coordinate:

$$\exists i \text{ st. } |\theta_i(0) - \theta'_i(0)| \leq \delta, \quad \text{and } \theta_j(0) = \theta'_j(0) \quad \forall j \neq i. \quad (6)$$

Let $\mathcal{M}$ denote a randomized consensus mechanism that maps initial states $\theta(0)$ to a sequence of messages and states observed during execution. $\mathcal{M}$ satisfies ϵ -DP if $\forall \delta$ -adjacent $\theta(0), \theta'(0)$ and measurable observations sets $\mathcal{S}$,

$$\Pr[\mathcal{M}(\theta(0)) \in \mathcal{S}] \leq e^{\epsilon\delta} \Pr[\mathcal{M}(\theta'(0)) \in \mathcal{S}]. \quad (7)$$

$\epsilon > 0$ is the privacy parameter, where smaller values correspond to stronger privacy guarantees. DP ensures that the observable behavior of the mechanism is statistically similar for adjacent initial states, limiting an adversary's ability to infer any single agent's state from the execution.

IV. SETUP

A. Baseline

We use the SPC framework from [1] as our baseline, where privacy is achieved by perturbing inter-agent messages during the consensus process. Instead of sharing exact state values, agents transmit noisy versions of their states at each iteration. The noise masks individual contributions while still allowing the system to converge to a common value.

At each round, agent i sends a privatized, noisy message

$$x_i(t) = \theta_i(t) + \eta_i(t), \quad (8)$$

where $\eta_i(t) \sim \text{Laplace}(cq^t)$ is drawn from a geometrically decaying Laplace distribution, where $c > 0$ controls the initial noise magnitude and $q \in (0, 1)$ is the rate of noise shrinkage over time. This ensures that early iterations provide stronger privacy and later ones allow for more accurate convergence.

In the centralized case, a server aggregates noisy messages:

$$y(t) = \frac{1}{N} \sum_{i=1}^N x_i(t), \quad (9)$$

and broadcasts this value back to all agents. Each agent then updates its state via interpolation:

$$\theta_i(t+1) = (1 - \sigma)\theta_i(t) + \sigma y(t), \quad (10)$$

where $\sigma \in (0, 1)$ controls the update step size.

The decentralized mechanism follows the same principle, with each agent averaging noisy messages from its neighbors instead of relying on a central server.

According to [1], the expected consensus is

$$\bar{\theta}(0) = \frac{\sum_{i=1}^N \gamma_i \theta_i(0)}{\sum_{i=1}^N \gamma_i} \quad \gamma_i = \frac{d_i + 1}{\sigma_i}. \quad (11)$$

Since all agents use the same update parameter ($\sigma_i = \sigma$), this reduces to the degree-weighted average. Throughout this work, we study undirected graphs and therefore evaluate convergence with respect to the degree-weighted average. This differs from standard decentralized consensus [2], where balanced undirected graphs converge to the simple average.

This mechanism achieves DP by ensuring that each transmitted message is obfuscated, while the geometric decay of noise preserves convergence in expectation. Thus, the system exhibits an inherent tradeoff between privacy (controlled by the noise parameters) and accuracy.

B. Implementation

We implement the SPC baseline from scratch in Python, as we could not find any publicly available codebase for reproducing the results in [1]. This required reconstructing the update rules, noise injection mechanism, and experimental setup from the original paper. Our implementation closely follows the theoretical formulation and serves as the foundation for all subsequent experiments and extensions.

Unless otherwise stated, initial states are independently sampled from $\text{Unif}(-3\sqrt{3}, 3\sqrt{3})$, where the interval is selected for numerical convenience. We evaluate the system using two primary metrics: accuracy and convergence. Accuracy is measured as the absolute difference between the converged consensus value and the degree-weighted average¹ of the initial states, as discussed in the previous section.

To evaluate convergence, we use the global potential function proposed in [1],

$$P(t) = \frac{1}{2} \sum_{(i,j) \in E} (\theta_i(t) - \theta_j(t))^2, \quad (12)$$

which quantifies disagreement between neighboring agents. The convergence time is defined as the smallest iteration t such that $P(t) < \lambda$, where λ is a fixed tolerance. Unless otherwise specified, we use $\lambda = 0.001$.

C. Privacy-Accuracy Tradeoff

To validate our implementation of the DP iterative consensus mechanism, we first reproduce the privacy-accuracy tradeoff observed in [1]. This serves as a baseline verification step before introducing any extensions.

¹The degree-weighted average of initial states is henceforth referred to as *true average* the system should converge to.

1) *Experimental Setup*: We consider the centralized SPC mechanism and vary the noise decay parameter $q \in (0, 1)$, *ceteris paribus*. As in [1], q directly controls the privacy level ϵ and the accuracy radius r . For each value of q , we compute:

- The theoretical privacy parameter ϵ
- The empirical accuracy radius r , defined as the deviation from the true average at convergence.

2) *Results*: Fig. 1 shows how ϵ and r move inversely as we vary q . As q increases, the magnitude of injected noise decays more slowly over time, resulting in stronger privacy guarantees (increasing ϵ). However, this improvement in privacy comes at the cost of reduced accuracy (increasing r).

Fig. 2 presents this tradeoff more directly by plotting the r as a function of ϵ . The results clearly demonstrate an inverse relationship: as privacy improves (smaller ϵ), the error bound r increases. This matches the theoretical characterization in [1], where the accuracy radius scales inversely with privacy.

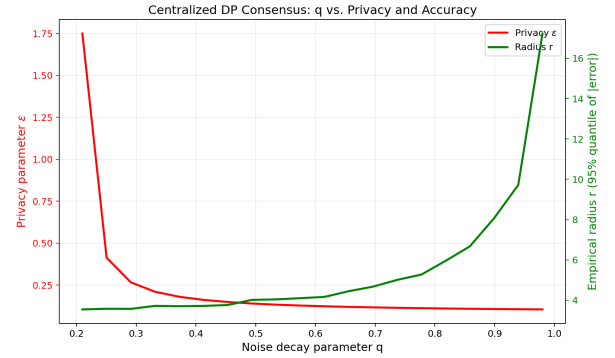


Fig. 1: Privacy parameter ϵ and accuracy radius r as functions of noise decay parameter q . As q increases, privacy improves (smaller ϵ), while accuracy degrades (larger r). For this graph, we draw initial states from $\text{Unif}(0, 1)$ for mathematical ease.

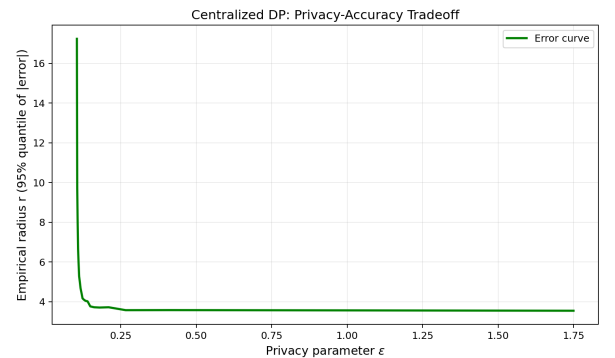


Fig. 2: Direct visualization of the privacy-accuracy tradeoff. Stronger privacy guarantees (lower ϵ) correspond to higher potential error in the final consensus value.

D. Extensions

We begin by evaluating the robustness of the baseline SPC mechanism under conditions not considered in the original

formulation. In particular, the baseline assumes reliable, noise-free communication and a fixed, uniform noise model across all agents. It does not account for communication imperfections or explore how alternative noise designs and network structure may impact performance. So, we first introduce a stress test that relaxes the assumption of perfect communication. We extend the communication model by incorporating additive Gaussian communication noise and packet loss. We then try to improve the performance, as measured by accuracy and convergence time, of the mechanism.

First, we study the impact of the DP noise distribution by replacing classical Laplace noise with Gaussian noise. For a fair comparison, we calibrate both mechanisms to achieve comparable privacy guarantees, allowing us to isolate the effect of the distribution on accuracy and convergence speed.

Second, we introduce a topology-aware noise mechanism. We no longer assume uniform noise injection across all agents, as in the baseline. In our approach, each agent independently scales its noise based on local structural information, specifically its degree in the communication graph. This preserves the decentralized nature of local DP while allowing more effective noise allocation.

Note that unless otherwise stated, we study the decentralized consensus mechanism due to its real-world relevance and more informative trade-offs in privacy, accuracy, and convergence.

V. STRESS TEST: LOSSY COMMUNICATION

A. Communication Noise

In addition to the privacy-preserving noise introduced by the Laplace mechanism, we study the effect of communication noise on decentralized consensus. Specifically, we augment the baseline Laplace DP mechanism by adding an additional Gaussian perturbation to each transmitted message, as in [15]. At each iteration t , agent i sends

$$x_i(t) = \theta_i(t) + \eta_i^{\text{DP}}(t) + \eta_i^{\text{comm}}(t), \quad (13)$$

where $\eta_i^{\text{DP}}(t) \sim \text{Laplace}(cq^t)$ is the privacy noise and $\eta_i^{\text{comm}}(t) \sim \mathcal{N}(0, \sigma_{\text{comm}}^2)$ models communication noise.

In the following experiments, we set $\sigma_{\text{comm}} = 0.1$. We choose a relatively small communication noise level to ensure the system exhibits meaningful convergence behavior, as larger noise levels can prevent convergence altogether under this metric. Across 1000 Monte Carlo trials, the baseline mechanism converges after 90.71 iterations on average, with a 95% confidence interval of [90.30, 91.12]. In contrast, under communication noise, none of the trials satisfy the consensus threshold within the simulation horizon (150 iterations). To enable a meaningful comparison of convergence behavior, we relax the convergence threshold to $\lambda = 0.1$. Under this, the system exhibits convergence within the simulation horizon, allowing us to compare convergence trends across settings.

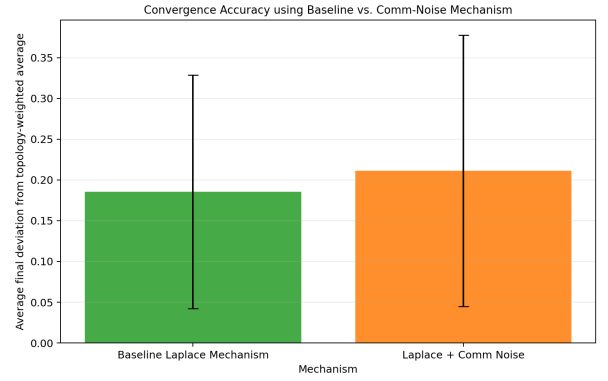


Fig. 3: Communication noise significantly increases the mean deviation from the true average. Note that large error bars primarily reflect variability induced by the DP and communication noise distributions across trials, rather than weak statistical significance.

Using this relaxed convergence threshold, the system with communication noise converges. As shown in Fig. 3, communication noise increases the steady-state error, with the mean deviation rising from 0.1839 (95% CI: [0.1531, 0.2146]) in the baseline case to 0.2016 (95% CI: [0.1681, 0.2351]) under communication noise, an approximate 9.63% increase.

This suggests that communication noise has a modest effect on steady-state accuracy but a significant impact on convergence under strict criteria. While the system fails to reach very low disagreement levels within the simulation horizon, it does exhibit convergence under a relaxed threshold, indicating that communication noise primarily slows the decay of disagreement rather than destabilizing the consensus process.

B. Packet Loss

We study the effect of lossy communication on DP consensus by introducing packet loss into the Laplace mechanism.

In the packet-loss variant, the noise model remains unchanged, but each directed message is independently dropped with probability p_{loss} , similar to [16]. As a result, each agent updates using a random subset of its neighbors at each round, reducing the effective information mixing in the network [17].

We measure convergence using the edge disagreement (potential) function. Across $n = 100$ Monte Carlo trials, the baseline mechanism reaches convergence after 90.71 iterations on average, with a 95% confidence interval of [90.30, 91.12]. Under packet loss with $p_{\text{loss}} = 0.5$, the average convergence time increases to 99.60 iterations, with a 95% confidence interval of [99.42, 99.78]. The non-overlapping confidence intervals indicate a statistically significant slowdown in convergence under packet loss.

Fig. 4 illustrates these effects. Packet loss leads to noisier trajectories and delayed consensus, as nodes repeatedly average over incomplete neighborhood information.

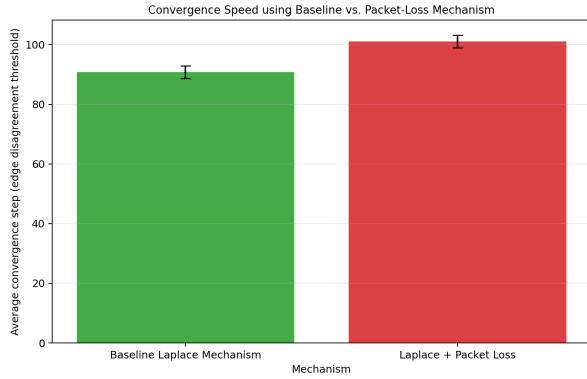


Fig. 4: A packet loss probability of Monte Carlo $p_{\text{loss}} = 0.5$ significantly increases the mean time to convergence.

In addition to slower convergence, packet loss increases the steady-state error. The mean deviation from the consensus value increases from 0.1839 (95% CI: [0.1531, 0.2146]) in the baseline case to 0.2007 (95% CI: [0.1684, 0.2330]) under packet loss, corresponding to an approximate 9.18% increase.

Overall, the SPC mechanism appears robust to lossy communication. Packet loss causes only a moderate slowdown in convergence and a small reduction in accuracy, even at high loss probabilities (Fig. 4). In contrast, communication noise has a much larger effect on convergence behavior. Under strict disagreement thresholds, noisy consensus often fails to converge within the simulation horizon, though convergence is recovered under relaxed thresholds with only modest accuracy degradation. These results suggest that communication noise impacts the mechanism more severely than packet loss.

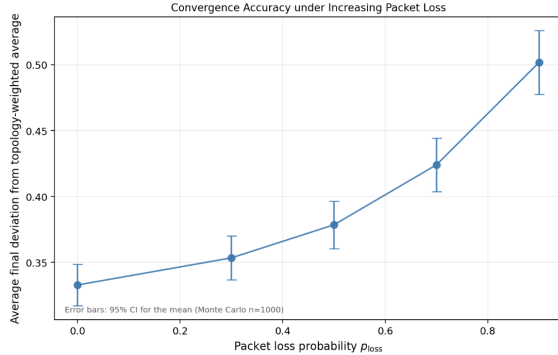


Fig. 5: Mean convergence error is increasing in p_{loss} .

VI. EXTENSIONS 1: GAUSSIAN DP

The baseline decentralized consensus mechanism in [1] achieves DP by adding geometrically decaying Laplace noise to agent messages. We replace the Laplace perturbations with Gaussian ones under the same protocol.

At each round t , agent i transmits

$$x_i(t) = \theta_i(t) + \eta_i(t), \quad \eta_i(t) \sim \mathcal{N}(0, \sigma_t^2), \quad \sigma_t = cq^t. \quad (14)$$

A. Probabilistic Privacy

To ensure a fair comparison between Laplace and Gaussian noise, we align the mechanisms under almost equivalent privacy guarantees. Both distributions are parameterized by their variance: Laplace(b) with $b = cq^t$, has $\text{Var}_{\text{lap}}(t) = 2b^2$, while $\mathcal{N}(0, \tau)$ directly gives $\text{Var}_{\text{norm}}(t) = \tau$.

We follow results in [7], which characterize Gaussian mechanisms under (ϵ, δ) -DP. Mechanism $\mathcal{M}$ satisfies (ϵ, δ) -DP if $\forall \delta$ -adjacent $\theta(0), \theta'(0)$ and measurable observations sets $\mathcal{S}$,

$$\Pr[\mathcal{M}(\theta(0)) \in \mathcal{S}] \leq e^{\epsilon\delta} \Pr[\mathcal{M}(\theta'(0)) \in \mathcal{S}] + \delta. \quad (15)$$

While the Laplace mechanism satisfies pure ϵ -DP, the Gaussian mechanism satisfies the relaxed notion of (ϵ, δ) -DP, where δ denotes a small probability of privacy failure due to the lighter Gaussian tails. In exchange, the smoother Gaussian noise may improve convergence behavior and consensus accuracy relative to Laplace noise.

We set δ arbitrarily small (1×10^{-5}) so that the Gaussian mechanism closely approximates ϵ -DP. Using Theorem 3.22 from [7], the Gaussian noise scale is calibrated according to

$$\tau = \frac{2\Delta_2(f)^2}{\epsilon^2} \ln\left(\frac{1.25}{\delta}\right), \quad (16)$$

where $\Delta_2(f)$ is ℓ_2 -sensitivity of function f . In consensus, the query f is the average over agent states, with ℓ_2 -sensitivity

$$\Delta_2(f) = \max_{\|x-x'\|_2 \leq 1} \left\| \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n x'_i \right\|_2 = \frac{1}{n}, \quad (17)$$

since values are drawn from $\text{Unif}(0,1)$ for analytic convenience. Substituting Eq. 17 into Eq. 16 yields the variance required for the Gaussian mechanism to match the same ϵ -privacy level as the Laplace mechanism.

This calibration ensures a fair comparison, isolating the effect of the noise distribution itself. Under this setup, we observe that Gaussian noise yields faster, more accurate, and more consistent consensus than Laplace noise.

B. Results

We evaluate both mechanisms over 200 Monte Carlo trials on a fixed Erdős-Rényi communication topology [18], with fixed initial states and parameters across all trials. All trials converged, with the distribution of times shown in Fig. 6. As predicted, the non-private mechanism converges the fastest because the agent values are not being perturbed by noise. The Gaussian mechanism converges over two times faster and with much lower variance than the Laplace mechanism.

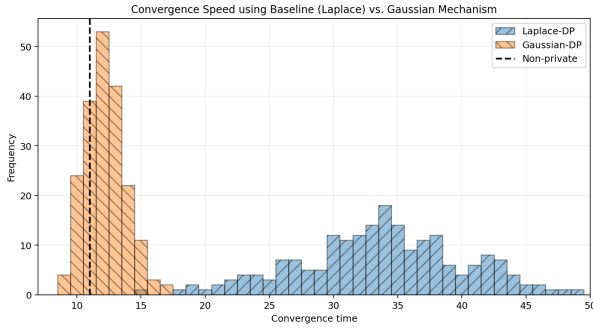


Fig. 6: Gaussian noise achieves faster and less variable convergence than Laplace noise

Using the same set of trials, we also evaluated the final consensus value against the true average. In Fig. 7, we find that both Laplace and Gaussian mechanisms are centered around the true mean, but the Gaussian mechanism exhibits much lower variance, improving accuracy compared to Laplace.

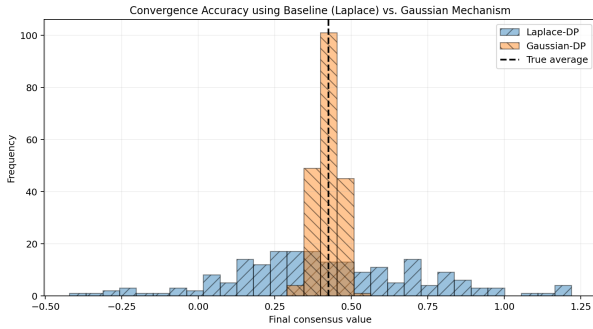


Fig. 7: Gaussian noise yields more accurate and consistent consensus than Laplace, with both centered near the true mean.

These results are consistent with the underlying noise distributions. Compared to Laplace noise, Gaussian noise has lighter tails and therefore produces fewer large perturbations to agent states during the consensus process. Thus, the Gaussian mechanism exhibits faster, more stable, and more accurate convergence across trials. This improvement comes at the cost of relaxing the privacy guarantee from pure ϵ -DP to (ϵ, δ) -DP, though the practical privacy levels remain comparable under the chosen parameters.

VII. EXTENSION 2: TOPOLOGY-AWARE DP NOISE

In the baseline mechanism, the DP noise added to each agent is drawn from the same distribution across all nodes,

$$\eta_i(t) \sim \text{Laplace}(cq^t). \quad (18)$$

While this provides uniform privacy, it ignores the heterogeneous influence of nodes. In particular, high-degree nodes propagate their updates to more neighbors and have a larger impact on the global state, yet they inject noise with the same scale as low-degree nodes [2] [19].

This leads to inefficient noise allocation. To address this, we introduce a topology-aware mechanism inspired by [20] that

scales the noise magnitude inversely with node connectivity, assigning less noise to more influential, high-degree nodes, as in [21]. Specifically, the noise added by agent i is drawn as

$$\eta_i(t) \sim \text{Laplace}\left(\frac{c'}{d_i + 1}q^t\right), \quad (19)$$

where d_i is the degree of node i ².

The inverse dependence on degree is a modeling choice that aligns noise magnitude with node influence. To ensure a fair comparison with the baseline, we select the global scaling factor c' so that the total injected noise variance is matched across mechanisms. This yields

$$c' = \frac{c}{\sqrt{\frac{1}{N} \sum_{i=1}^N \frac{1}{(d_i + 1)^2}}}. \quad (20)$$

The denominator contains $\frac{1}{N} \sum_{i=1}^N (d_i + 1)^{-2}$, the *average inverse squared degree*. This quantity is small in networks with many high-degree nodes, since $(d_i + 1)^{-2}$ decreases rapidly with degree, and larger in sparse networks where degrees are small. Accordingly, c' increases in dense networks and decreases in sparse ones. This is consistent with inverse-degree scaling: in dense networks, many nodes have large d_i , so the per-node noise is heavily attenuated, requiring a larger global scaling factor c' to preserve the total variance.

Notably, because $(d_i + 1)^{-2}$ is a convex, decreasing function of d_i , the average is disproportionately influenced by low-degree nodes. As a result, the normalization is more sensitive to sparsely connected parts of the network, even though the noise allocation itself reduces noise at highly connected nodes.

A. Heterogeneous Privacy

Unlike the baseline mechanism, which provides a uniform privacy level across nodes, the topology-aware mechanisms provides heterogeneous privacy guarantees. Each node i effectively uses its own noise scale $c_i = \frac{c'}{d_i + 1}$, and can therefore be associated with a node-specific privacy parameter as in [1]:

$$\epsilon_i = \frac{q}{c_i(q + \sigma_m - 1)} \propto \frac{1}{c_i}. \quad (21)$$

Higher-degree nodes (with larger d_i) have smaller c_i , receive less noise, and incur larger privacy loss (larger ϵ_i), while lower-degree nodes receive stronger privacy. As a result, the mechanism no longer provides a uniform ϵ -DP guarantee across nodes. The overall (worst-case) privacy level is governed by the largest ϵ_i , which corresponds to the highest-degree node, and is therefore weaker than the baseline guarantee for the same nominal parameter c . In particular,

$$\epsilon_{\max} = \max_i \epsilon_i = \frac{q}{(q + \sigma_m - 1)} \cdot \frac{d_{\max} + 1}{c'}, \quad (22)$$

where $d_{\max} = \max_i d_i$. Thus, the proposed approach trades uniform privacy for improved performance by reallocating noise according to node influence while preserving total variance.

²The +1 accounts for an implicit self-loop, since each node updates by interpolating between its own value and those of its neighbors. This ensures that $d_i + 1$ reflects total influence including self-contribution.

1) *Proof:* We now show that the choice of c' in Eq. 20 ensures that the total noise variance injected at each round t is equal across both mechanisms.

In the baseline mechanism, each agent adds noise

$$\eta_i(t) \sim \text{Laplace}(cq^t), \quad (23)$$

which has variance $2c^2q^{2t}$. Summing over all N agents,

$$\text{Var}_{\text{baseline}}(t) = 2Nc^2q^{2t}. \quad (24)$$

In the topology-aware mechanism, noise is scaled by degree:

$$\eta_i(t) \sim \text{Laplace}\left(\frac{c'}{d_i + 1}q^t\right), \quad (25)$$

so agent i contributes variance $2(c')^2q^{2t}(d_i + 1)^{-2}$. Summing across agents,

$$\text{Var}_{\text{topology-aware}}(t) = 2(c')^2q^{2t} \sum_{i=1}^N \frac{1}{(d_i + 1)^2}. \quad (26)$$

Equating total variance,

$$2Nc^2q^{2t} = 2(c')^2q^{2t} \sum_{i=1}^N \frac{1}{(d_i + 1)^2}, \quad (27)$$

yields

$$c' = c \cdot \sqrt{\frac{N}{\sum_{i=1}^N \frac{1}{(d_i + 1)^2}}}. \quad (28)$$

which simplifies to

$$c' = \frac{c}{\sqrt{\frac{1}{N} \sum_{i=1}^N \frac{1}{(d_i + 1)^2}}}. \quad (29)$$

The factor q^{2t} cancels, so the scaling depends only on the network topology, not temporal decay.

B. Results

Topology-aware scaling has negligible impact on Erdős–Rényi graphs, but on heterogeneous-degree star and Barabási–Albert graphs [22], it reduces final error and variance while preserving convergence rates.

For example, in star graph topology, the average steps for convergence go from 88.885 (95% CI: [88.564, 89.206]) to 87.680 (95% CI: [87.401, 87.959]). In regular topology, the time to convergence remains exactly the same at 94.915 (95% CI: [94.570, 95.260]).

In terms of deviation from the average, Figures 8 and 9 show these effects for representative heterogeneous graphs. In both cases, the topology-aware mechanism yields a lower mean final error compared to the baseline, as shown by the difference in the bar heights. The error bars also indicate reduced variability across trials, suggesting more stable convergence behavior when noise is scaled by node degree. These results suggest that topology-aware noise allocation improves consensus accuracy and stability in heterogeneous networks by reducing noise at high-degree nodes.

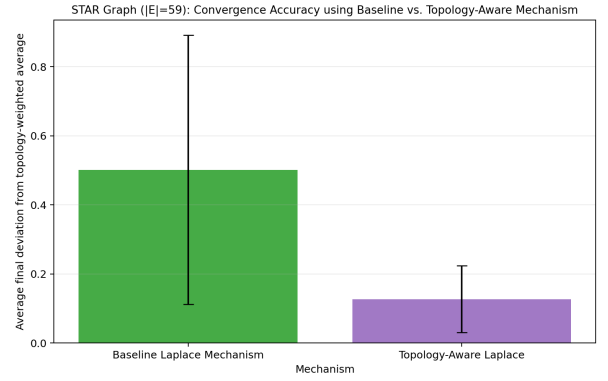


Fig. 8: On star graphs, the topology-aware mechanism converges more accurately than the baseline.

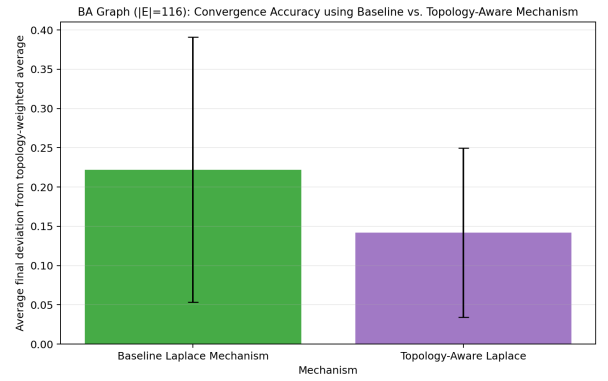


Fig. 9: On Barabási–Albert graphs, the topology-aware mechanism converges more accurately than the baseline.

In heterogeneous graphs, high-degree nodes dominate information flow, so reducing noise at these nodes significantly improves accuracy. In contrast, in homogeneous graphs where node influence is uniform, topology-aware scaling provides little benefit over the baseline.

VIII. TRADEOFFS

A. Compute Cost of Laplace vs. Gaussian Sampling

The Laplace mechanism is computationally simpler to implement due to its closed-form inverse cumulative distribution function (CDF), which enables direct inverse-transform sampling. In contrast, without a closed-form inverse CDF, Gaussian sampling typically requires more complicated methods such as Box–Muller [23] or Ziggurat [24].

However, in our setting, this distinction is negligible. Our implementation relies on optimized numerical libraries, which provide efficient, high-quality sampling for both distributions. As a result, while Laplace noise is theoretically simpler, sampling complexity does not meaningfully impact the empirical performance or implementation of our system.

On real hardware, Gaussian sampling has a higher computational overhead than Laplace sampling. In resource-constrained or latency-sensitive deployments, this overhead

can become a relevant systems-level tradeoff. Nevertheless, our empirical results indicate that Gaussian noise achieves higher accuracy and faster convergence than Laplace noise under comparable privacy budgets, which may justify the increased sampling cost in performance-critical applications.

B. Probabilistic Gaussian DP Guarantee

A key tradeoff in using the Gaussian mechanism is that it provides (ϵ, δ) -DP rather than pure ϵ -DP of the Laplace mechanism, introducing a small failure probability $\delta > 0$ in the privacy guarantee. While δ can be chosen to be arbitrarily small (e.g., $\delta = 10^{-5}$ in our experiments), this still represents a relaxation of the strict guarantees provided by pure ϵ -DP.

This highlights a fundamental tradeoff: the Gaussian mechanism achieves improved empirical performance (faster, more accurate convergence with less variance) at the cost of a weaker privacy guarantee. In contrast, the Laplace mechanism provides strict ϵ -DP but has heavier-tailed noise, which leads to higher variability and slower convergence.

C. Topology-Aware Privacy Tradeoff

A key tradeoff of the topology-aware mechanism is the introduction of *heterogeneous privacy*. In baseline SPC, all nodes inject identically distributed noise, yielding a uniform ϵ -DP guarantee. In contrast, the topology-aware mechanism scales noise inversely with node degree, producing node-specific privacy levels ϵ_i . Consequently, high-degree nodes inject less noise and therefore receive weaker privacy protection than lower-degree nodes.

As a result, the mechanism no longer satisfies a single global ϵ -DP guarantee; privacy is instead governed by the worst-case parameter $\epsilon_{\max} = \max_i \epsilon_i$, which exceeds the baseline ϵ under equal total noise variance. This tradeoff may be undesirable in some deployments, since high-degree nodes are often more central, influential, or information-rich participants in the network. Thus, the topology-aware mechanism improves performance, but at the cost of uneven and potentially riskier privacy guarantees across agents.

IX. CONCLUSIONS & FUTURE DIRECTIONS

In this work, we evaluated the robustness of the SPC framework under more realistic conditions, including lossy communication, alternative noise distributions, and topology-aware noise design. Our goal was to understand how these extensions affect convergence behavior and accuracy beyond the assumptions of the baseline model.

We validate the SPC framework through simulation, confirming its theoretical guarantees on DP, convergence, and the privacy–accuracy tradeoff, which are not extensively evaluated empirically in [1]. Our results show consistent convergence and agreement with theoretical predictions, establishing a reliable empirical baseline for evaluating our extensions.

Building on this baseline, we first evaluated the robustness of SPC by introducing communication failures, namely communication noise and packet loss. These served as a stress test to assess how the baseline behaves under realistic conditions.

We find that persistent perturbations, such as additive communication noise, have the largest impact, significantly slowing convergence and increasing final error. In contrast, packet loss produces a more moderate effect: even at relatively high loss rates, the system continues to converge in nearly all runs, albeit more slowly and slightly less accurately. This suggests that SPC is inherently more sensitive to continuous noise injection than to intermittent loss of information.

We then extended the mechanisms to try to improve performance, specifically through testing an alternative noise distribution and a topology-aware design. These extensions move beyond robustness to explore how modifying the noise model and leveraging graph structure can improve performance.

We found that Gaussian noise leads to faster and more stable convergence compared to Laplace under comparable privacy settings, indicating that tail behavior affects system dynamics. Additionally, topology-aware noise improves performance in heterogeneous graphs by reducing noise at high-degree nodes, yielding lower error and variance, while providing negligible gains in homogeneous-degree graphs.

There are several directions for future work. First, all results presented here are empirical, motivating a theoretical analysis of the observed privacy–accuracy–convergence tradeoffs. In particular, it would be great to formally characterize how alternative noise distributions and topology-aware noise scaling affect convergence rate, steady-state error, and DP guarantees.

We would also like to explore more alternative noise distributions, like the truncated Laplace and Gaussian. Our results on the Gaussian mechanism show that the choice of noise distribution significantly affects consensus dynamics even under comparable privacy guarantees. Distributions with lighter or truncated tails, as well as hybrid time-varying mechanisms, may further improve convergence speed and accuracy.

For topology-aware noise, future work should establish formal guarantees relating graph structure to performance gains in heterogeneous networks. We also plan to develop a Byzantine-robust extension of this mechanism. Because topology-aware noise reduces perturbation at high-degree nodes, these influential nodes become attractive attack targets. One approach is to incorporate lightweight Byzantine filtering based on expected state ranges: if initial states lie within a known bounded interval, confidence bounds on noisy updates can identify and exclude implausible reports. While adaptive adversaries may evade detection by selecting less extreme deviations, this limits their impact. A key challenge is avoiding false positives due to legitimate outliers or large noise realizations.

Another promising direction is the study of hierarchical or fog-based architectures [28], where agents communicate through trusted local coordinators or hubs. Such structures may enable more efficient noise allocation and scalable privacy-preserving consensus in large networks.

Overall, this work shows that SPC remains robust under realistic communication conditions and that incorporating network structure and noise design can substantially improve practical performance while preserving privacy.

CODE AVAILABILITY

The code used to reproduce the experiments and results in this paper is publicly available at:

https://github.com/vihaann06/dp_for_consensus

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